



Multi-frequency asynchronous optical coherence elastography by direct demodulation

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Abstract: Optical coherence elastography has demonstrated promising results in highly specialized, research-grade OCT systems and in systems with custom, slow scan patterns. However, these approaches are fundamentally incompatible with commercial, FDA-approved OCT systems already deployed in clinics worldwide. Asynchronous optical coherence elastography addresses this limitation by using raster scanning–induced amplitude modulation to spatially encode harmonic shear wave displacement fields. This principle enables recovery of the coherent shear wave field within seconds using conventional frame-rate systems, despite severe temporal undersampling across B-scans. In this work, we extend this approach to show that asynchronous imaging can recover multiple simultaneous excitation frequencies, effectively multiplexing the determination of shear wave dispersion curves. We further introduce a new recovery framework, termed *direct demodulation*, based on the two-dimensional spatial Fourier transform of raster-scanned displacement data. We demonstrate experimentally that this framework enables efficient and intuitive isolation of up to 12 excitation frequencies within a single acquisition. We also identify practical limits on frequency multiplexing and provide guidelines and code for selecting imaging and excitation parameters that minimize spectral interference. The framework is validated through agreement between sequential and multiplexed measurements of shear wave number. Finally, we demonstrate the importance of multi-frequency elastography for robust mechanical characterization of complex tissues, showing sensitivity to boundary effects, viscoelastic dispersion, and improved mechanical contrast through incoherent frequency averaging in heterogeneous samples. This work enables multi-frequency, shear wave–based OCE in conventional frame-rate, raster-scanning OCT systems, further paving a path toward widespread adoption of elastography in existing imaging platforms.

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1. Introduction

Optical coherence tomography (OCT) is a high-resolution imaging modality that uses interferometry to detect backscattered light from a sample. It is capable of producing depth-resolved, cross-sectional images with micrometer-scale resolution and millimeter-scale penetration depth [1,2]. Since its inception, OCT has achieved widespread clinical adoption, with over 30 million procedures performed annually in ophthalmology [3] and over 100,000 coronary imaging procedures performed each year to guide interventions for coronary artery disease [4,5]. Although conventional OCT systems are largely limited to structural imaging, continued interest has remained in advancing its capabilities to enable mechanical characterization via elastography because alterations in tissue elasticity are a hallmark of numerous disease processes, including cancer, pulmonary fibrosis, and corneal ectasias [6–8].

Optical coherence elastography (OCE) is an extension of OCT which provides measurements of tissue elasticity [9–11]. By using OCT to track tissue displacements under external excitation,

an appropriate constitutive model can reveal estimates of mechanical properties such as shear modulus or Young's modulus. Among various emerging OCE methods, shear wave elastography is a promising, dynamic approach which measures the size and speed of propagating shear waves within the sample of interest [12–15]. Despite its potential, widespread translation of shear wave OCE has remained challenging due to the mismatch between rapidly propagating shear waves and relatively slow OCT frame-rates leading to temporal undersampling. To address this, most implementations of shear wave OCE rely on phase-locked, or "MB-mode" scanning protocols, in which shear wave excitation is repeated thousands of times (e.g., 10,000 repetitions for a 100×100 grid) while the OCT beam remains stationary at each location within the grid. Although effective *ex vivo*, this approach is susceptible to motion artifacts and results in long acquisition times. Ultrafast MHz lasers have been proposed and demonstrated to improve imaging speed [16–19], but these systems are highly specialized for elastography at the expense of structural image quality, often yielding low signal-to-noise ratios and limited axial range.

Previously, we developed asynchronous OCE (AsyncOCE), a harmonic shear wave OCE technique which eliminates the need for phase-locked imaging. AsyncOCE reconstructs the spatially and temporally coherent shear wave field in three dimensions directly from continuously acquired, raster-scanned B-scans through signal processing. It is also compatible with standard frame-rate, phase-stable OCT systems, which are already in clinics around the world. In addition to being at least three times faster than comparable MB-mode methods, AsyncOCE allows for a 15-fold increase in lateral sampling, providing high quality structural image sampling and enabling advanced motion correction techniques. A more comprehensive comparison between AsyncOCE and MB-mode imaging is provided in our previous work [20,21].

Despite these advantages, AsyncOCE has only been demonstrated with single-frequency shear wave excitation. This limits our ability to characterize the frequency-dependent, viscoelastic properties typical of most biological tissues [22]. Multi-frequency analysis is also essential for characterizing tissues with significant boundary discontinuities, such as the cornea, where shear wavelength is on the same scale as sample thickness [23]. Furthermore, multi-frequency analysis can be leveraged to overcome amplitude nulls, which occur from destructive interference of harmonic shear waves. Just as incoherent averaging of multiple OCT frames can reduce speckle noise, integrating information from multiple shear wave frequencies has been shown to improve elastography image quality and reveal contrast from smaller features that single-frequency measurements may otherwise miss [24]. To more accurately assess the mechanical characteristics of complex tissue across a variety of conditions, it is therefore highly desirable to perform elastography measurements across a range of many wave excitation frequencies.

It is relevant to note that, in contrast to harmonic shear wave methods such as AsyncOCE, impulse-excitation of shear waves already inherently excites a broadband response, enabling simultaneous access to a wide range of shear wave frequencies [19]. However, these approaches typically launch and measure shear waves along just a single direction, which may make them susceptible to interference and reflections at tissue boundaries, particularly in heterogeneous structures. Characterization of laterally anisotropic tissue samples therefore requires measurements to be performed along each direction independently [25]. Due to the broadband frequencies that are excited in the sample, most implementations also continue to rely on MB-mode scanning, custom full-field OCT systems, or ultrafast laser systems, all of which complicate clinical translation. From a practical standpoint, generating a tightly focused point of impulse excitation further complicates translation in deep organs such as the coronary arteries. Given these considerations, and our emphasis on compatibility with existing clinical OCT systems, we focused our attention in this work on extending AsyncOCE to incorporate multi-frequency measurements of harmonic waves.

One direct approach to extending AsyncOCE to multi-frequency measurements would be to measure each frequency sequentially, one after the other. However, imaging time would

scale linearly with the number of frequencies sampled. This becomes a critical limitation for applications *in vivo*, where acquiring a single volume over a 12 by 12 mm field of view requires approximately 2 seconds. Extending this to just five frequencies, for example, would increase acquisition time to 10 seconds, which in certain applications (such as the eye's anterior segment) represents an acquisition time too long for practical imaging in patients.

Alternatively, multiple frequencies could be simultaneously excited in the sample, such as in Asemani et al. [26]. Then, a Fourier transform along time, coupled with an appropriate bandpass filter around each frequency, could separate the signals. However, this would require Nyquist sampling of at least twice the highest frequency of interest. For 2 kHz shear wave excitation, a minimum 4 kHz frame rate would be required; conventional OCT frame rates, however, are more than two orders of magnitude slower. Although phase-locked, MB-mode imaging can virtually reconstruct shear wave fields with an effective frame rate equivalent to the OCT A-line rate, this requires the use of excitation frequencies for which an integer number of shear wave periods fit exactly inside the M-mode scan for each location; this impedes dense frequency sampling in a narrow bandwidth. To overcome this limitation, Asemani et al. used a 50% imaging duty cycle for each M-mode, stopping and restarting the excitation for each new lateral scan location. Ultrafast lasers and fixed-frequency resonant scanners can also directly provide frame rates up to 8 kHz. However, this work focuses on enabling multi-frequency measurements acquired with a commercial frame-rate OCT system.

AsyncOCE relies on amplitude modulation, so it directly enables simultaneous multiplexing of shear wave frequencies within a single excitation event. Since there is a non-negligible time delay between each sample acquired within an OCT volume, frequencies become encoded as a function of *space* rather than time, allowing many frequencies to be encoded, despite relatively slow frame rates. In this work, we establish the theoretical framework for multi-frequency excitation and demodulation. We also introduce a new general framework for AsyncOCE, called *direct demodulation*. In contrast to *split-axis demodulation* introduced previously [21], which recovers the shear wave field along each scanning axis individually, direct demodulation directly recovers the shear wave field from the 2D spatial Fourier transform of the raster-scanned shear wave field. Direct demodulation also straightforwardly reveals the ability to simultaneously encode 12 shear wave frequencies in the biomedically relevant frequency range (500 Hz–3 kHz), efficiently utilizing the available support in the lateral power spectrum of conventional Doppler OCT imaging. We also lay out the practical limitations on frequency multiplexing, such as how many frequencies may be simultaneously encoded before interference occurs, and how imaging parameters should be chosen to reduce interference. Sample code is provided to enable selection of optimal scanning and excitation parameters, as well as to perform multi-frequency AsyncOCE. This approach was validated by demonstrating strong agreement between sequential and multiplexed measurements of shear wave number, as well as both direct and split-axis demodulation. We further highlight the importance of multi-frequency analyses by showing how boundary effects and viscoelastic properties in tissue-mimicking phantoms can impact wave speed dispersion. Finally, we demonstrate that incoherent averaging across multi-frequency shear wave fields enhances elastography image contrast in a heterogeneous sample, emphasizing the importance of multi-frequency measurements for robust analysis of complex tissues.

2. Methods

First, we describe split-axis demodulation, following the approach previously outlined in Schmidt et al. [21], but with reformulated expressions to better highlight underlying principles. Next, we introduce our new demodulation strategy for AsyncOCE, which we term direct demodulation. This method achieves equivalent results, while providing a clearer interpretation in Fourier space. The direct demodulation framework naturally supports multi-frequency AsyncOCE, enabling straightforward extension to multiple excitation frequencies. We also detail the experimental

setup, including materials, OCE system configuration, signal processing pipeline, modeling methods, and the frequency-averaging strategy used to combine results across different excitation frequencies.

2.1. Split-axis demodulation AsyncOCE

A right-handed coordinate system is defined with depth increasing along the optical axis z . OCT acquires one-dimensional depth profiles along z , known as A-lines. To construct a 2D image, the laser beam is scanned laterally along x to produce a B-scan [see Fig. 1(a)]. A 3D volume is finally obtained by acquiring successive B-scans along the out-of-plane lateral dimension y .

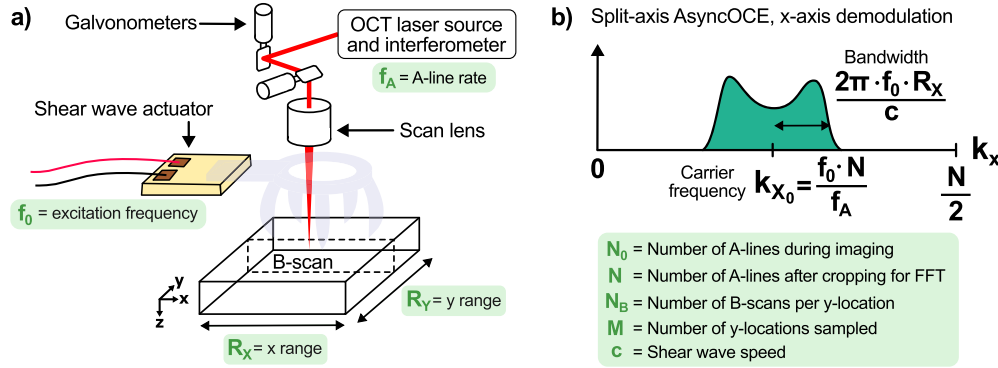


Fig. 1. (a) OCT system and key parameters are defined. Conventional OCT systems use galvanometers to scan and form volumetric images. Depth-profiles along z are acquired by the OCT imaging beam. The beam is scanned along x to form a B-scan. B-scans are acquired along y to form a three dimensional volume. The shear wave actuator may be driven by one or more frequencies. (b) An illustrated shear wave displacement power spectrum computed along the B-scanning dimension x is shown. In split-axis demodulation, we first center this signal on 0 Hz, then apply a low-pass filter, then perform demodulation independently along the y -axis.

Discrete A-line positions within a B-scan are denoted by x_n , where $n \in \{1, \dots, N\}$, and N is the total number of A-lines per B-scan. Similarly, each discrete B-scan position is represented by y_m , where $m \in \{1, \dots, M\}$, and M is total number of locations sampled along the y -dimension. In elastography, each A-line location must be sampled at least twice for relative displacement measurements. The number of repeat measurements per A-line location is given by N_B . During raster-scanning, N_B is equivalent to the number of repeat B-scans per y_m . We account for incomplete imaging duty cycle, which can arise from galvanometer flyback patterns and cropping, by designating the raw number of A-lines per B-scan as N_0 .

Because shear wave propagation speed is on the order of meters per second—much faster than typical OCT frame rates—directly measuring shear waves from raster-scanned data causes severe temporal undersampling. Rather than neglecting the time delay between neighboring A-lines from within a B-scan, we can account for it explicitly with the phase term

$$\phi[x_n] = e^{j2\pi f_0 \Delta t_a x_n}, \quad (1)$$

where f_0 is the shear wave excitation frequency and Δt_a is the time delay between A-lines. As x_n increases over the course of a B-scan, the time delay accumulates linearly as a function of space. Given laser wavelength-sweep rate f_A , the time delay between A-lines is $\Delta t_a = 1/f_A$. Similarly,

we can account for the time delay along the y-axis between out-of-plane B-scans with

$$\phi[y_m] = e^{j2\pi f_0 \Delta t_b y_m}, \quad (2)$$

where $\Delta t_b = N_0 N_B / f_A$. Note the usage of N_0 , the raw number of A-lines per B-scan acquired during imaging, which is distinct from N , the number of A-lines in the tomogram after post-processing. Although some A-lines may be cropped out or excluded due to galvanometer down time, their impact on the time delay between B-scans must still be accounted for. Since $\Delta t_x x_n$ and $\Delta t_y y_m$ both increase linearly along x and y , respectively, their phase delays also accumulate linearly along both axes as a function of space. We refer to these as “phase ramps”.

Previously, raster scanning was described as an amplitude modulation of the displacement field, which shifts the desired signal to a carrier frequency in the frequency domain [21]. This interpretation is consistent with the above phase-ramp representations of raster-scanning. They are unified by the Fourier shifting property, which states that a shift in one domain corresponds to a phase ramp in the other domain. In the context of shear wave elastography, a phase ramp in the spatial domain corresponds to a frequency shift in the Fourier k -space.

To counteract raster-scanning effects, the phase ramps induced by raster-scanning can be directly canceled by applying an opposing phase term. In k -space, applying an opposing phase term shifts the desired signal to 0 Hz (DC), which is then low-pass filtered to demodulate the effects of raster-scanning. This procedure results in a complex-valued, analytical displacement field. Despite originating from real-valued Doppler displacements, the demodulation procedure has a side effect analogous to the Hilbert transform. We can also define carrier frequencies k_{X0} and k_{Y0} by their pixel value along the k_x and k_y axes, respectively:

$$k_{X0} = f_0 \Delta t_a N = f_0 \frac{1}{f_A} N, \quad (3)$$

$$k_{Y0} = f_0 \Delta t_b M = f_0 \frac{N_B N_0}{f_A} M. \quad (4)$$

N and M are the number of samples along x and y , respectively, used for Fourier transformation after tomogram cropping. Figure 1(b) shows an illustration of the displacement power spectrum along k_x where the shear wave field signal has been amplitude modulated to k_{X0} due to raster-scanning. Because the raw displacement signal is real-valued, its spectrum is symmetric about 0 Hz, so only the positive frequencies are shown. Given this visual representation of raster-scanned data, one may also intuit how an opposing phase term should be applied along each scanning axis to re-center, or “downshift”, the signal from k_{X0} back to 0 Hz. The discrete Fourier transform shift property conveniently enables such non-integer shifting. If S_0 is the raw displacement acquired by raster-scanning, and S_{dx} is the demodulated displacement field along x , then we can write this as

$$S_{dx}[x_n, t_i] = \mathcal{L}_{k_x} \left\{ S_0[x_n, t_i] \cdot \exp \left(-j2\pi f_0 \frac{1}{f_A} x_n \right) \right\}, \quad (5)$$

where $\mathcal{L}$ denotes a low-pass filter that suppresses the mirror copy of the signal at the opposite carrier frequency, while also reducing noise and producing the desired complex-valued, analytical displacement signal. After demodulation along the x -axis, a second phase ramp and low pass filter are applied independently along the y -axis to account for the cumulative time delay between out-of-plane B-scan locations:

$$S_{dxy}[y_m, t_i] = \mathcal{L}_{k_y} \left\{ S_{dx}[y_m, t_i] \cdot \exp \left(-j2\pi f_0 \frac{N_0 N_B}{f_A} y_m \right) \right\}. \quad (6)$$

Since demodulation is performed independently along each axis, we refer to this strategy as *split-axis demodulation*.

2.2. Direct demodulation AsyncOCE

Rather than performing separate demodulation and low-pass filtering steps along each axis, we introduce a new technique called direct demodulation. In this approach, both phase ramps (or equivalently, Fourier-domain shifts) are applied consecutively *before* any low-pass filtering. The shift amounts are functions of the time delays between samples and are the same as those defined previously in Eqs. (1) and (2). The key difference is that the 2D low-pass filter is applied simultaneously across both dimensions only *after* shifting along both k_x and k_y . Mathematically, this can be written as

$$S_d[x_n, y_m, t_i] = \mathcal{L}_{k_x, k_y} \left\{ S_0[x_n, y_m, t_i] \cdot \exp \left(-j2\pi f_0 \frac{1}{f_A} x_n \right) \cdot \exp \left(-j2\pi f_0 \frac{N_0 N_B}{f_A} y_m \right) \right\}. \quad (7)$$

Figure 2(a) illustrates the 2D spatial Fourier transform of a single-frequency shear wave field prior to demodulation. In a homogeneous sample with a diffuse shear wave field, the displacement signal is a ring in Fourier space, where the radius corresponds to the shear wave number. Raster-scanning shifts this ring to carrier frequencies k_{x_0} and k_{y_0} . These carrier frequencies are equivalent to those defined in Eqs. (3) and (4). To demodulate the effect of raster-scanning, re-center the ring at 0 Hz by applying phase ramps along both x and y in the spatial domain. A 2D low-pass filter, such as a Gaussian, is applied afterwards to suppress noise and the complex-conjugate copy of the signal. This process yields the complex-valued, demodulated, three-dimensional displacement field. Implementation and signal processing details follow in Section 2.6.

In split-axis demodulation, we previously only observed the projection of the field onto the k_x axis [see Fig. 2]. As a result, multi-frequency imaging appeared feasible only along a single dimension, limiting the number of frequencies that could be simultaneously multiplexed without interference. In contrast, direct demodulation makes it straightforward to visualize how the full 2D power spectrum of the displacement signal can be maximally exploited for multi-frequency imaging.

2.3. Extension to multi-frequency AsyncOCE via direct demodulation

In multi-frequency AsyncOCE, several frequencies are simultaneously driven in the sample. Rather than a single excitation frequency f_0 , we define multiple excitation frequencies by f_s , with f_1 being the lowest frequency. Each frequency is proportional to its carrier frequency, resulting in separation of the displacement fields in k -space. For example, six simultaneous, amplitude-modulated displacement fields are illustrated in Fig. 2(b). To demodulate each frequency, the direct demodulation pipeline may be repeated six times, once for each f_s , taking care to choose an appropriate low-pass filter width to preserve the full signal while suppressing other frequencies.

Although the signal processing pipeline is, in theory, a straightforward extension of single-frequency AsyncOCE methods, careful attention must be given to excitation frequency selection and imaging system parameters prior to data acquisition. For example, if carrier frequencies are too close to one another, or the bandwidths are too wide, signals could interfere and lead to inaccurate measurements. Given the importance of frequency selection, we provide a script which generates suggested frequencies in [Code 1](#) (Ref. [27]). The script also enables simulation of how key parameters influence the k -space distribution of carrier frequencies.

To better understand the factors that influence spatial bandwidth in the k -space, we define them here as a function of several key system and sample parameters. First, the shear wave number for a given shear wave speed c_s is $k_s = 2\pi f_s / c_s$. The shear wave number defines the radius of each ring, which is important to account for when aiming to preventing spectral overlap. In the

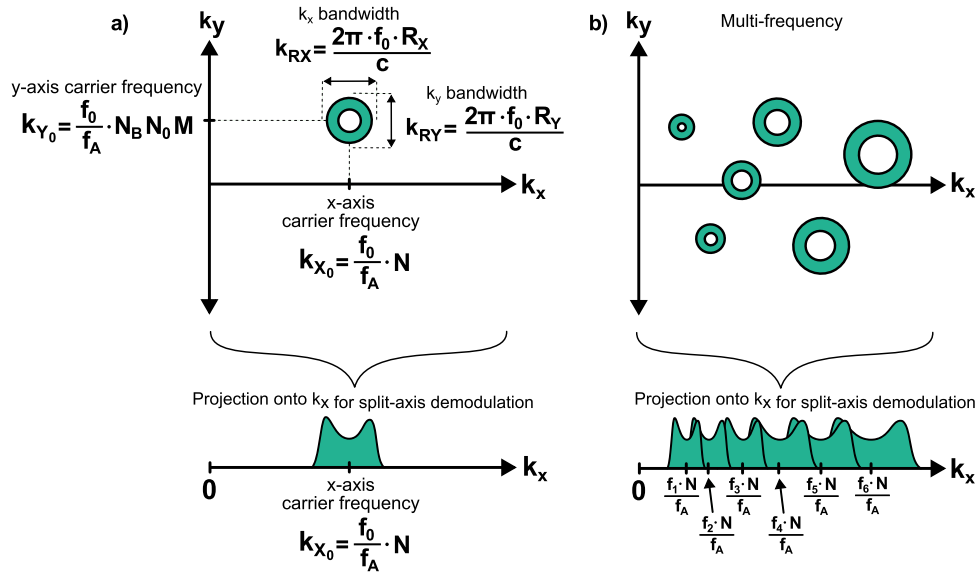


Fig. 2. (a) An illustration of the 2D spatial Fourier transform of the Doppler OCT displacement signal for a single excitation shear wave frequency along the x and y -scanning axes is shown. The signal is amplitude modulated along *both* k_x and k_y . The radius of the signal is related to the shear wave number. If the x and y scan ranges are uniform and the sample is homogeneous, the signal will be a circle. For anisotropic scan ranges, the circle will be an oval. Previously, split-axis demodulation only visualized the projection of this signal onto the k_x axis. (b) We can directly extend this framework to multi-frequency measurements. Larger frequency shear waves will have larger shear wave number, resulting in larger diameter signals. In split-axis demodulation, the projection onto the k_x -axis would appear to result in interference between frequencies. However, since the signal is actually modulated along both axes, we can safely measure many more than initially may be expected.

discrete k -space, divide by $dk_x = 1/R_X$ and $dk_y = 1/R_Y$ to convert the shear wave number into pixel units. This gives the exact ring radius (or bandwidth) in pixels:

$$k_{RX} = 2\pi f_s R_X / c_s, \quad (8)$$

$$k_{RY} = 2\pi f_s R_Y / c_s. \quad (9)$$

The radius of each ring will be smaller at lower excitation frequencies for a linear elastic material. This allows for denser selection of excitation frequencies at lower frequencies. At higher frequencies, the radius will be larger for the same shear wave speed, so excitation frequencies may need to be chosen more sparsely. We also consider this phenomenon intuitively: at low frequencies, shear waves tend to have a larger wavelength; at high frequencies, shear waves will have a smaller wavelength. Given the same field of view R , the spatial frequency content of lower frequency waves will be lower, resulting in a smaller radius in the k -space.

Reducing the B-scan range, or cropping the image and reducing R , will also reduce radii across all frequencies. We note from Eqs. (3) and (4) that the carrier frequencies are proportional to the number of samples N and M , so denser sampling will result in a larger spread between each carrier frequency while k_{RX} and k_{RY} remain constant.

Shear wave speed is intrinsic to a sample and typically beyond our control. However, it also plays a key role in determining the radius of the ring, and equivalently its bandwidth, around a carrier frequency. Stiffer samples with faster shear wave speeds c_s produce smaller rings,

whereas softer samples produce larger rings. Therefore, for heterogeneous samples, frequency selection should be optimized for the softest expected material. The sample script provided for frequency selection [27] accounts for this by allowing users to define the minimum and maximum expected frequencies of a viscoelastic material within the desired frequency range. In general, for a given clinical application, a range of possible shear moduli is known or can be determined in initial testing. For instance, for a new application with an unknown shear modulus range, single-frequency or sparse multi-frequency imaging could first be performed to estimate the expected range of shear wave speeds. A safety factor could then be applied to the lowest observed shear wave speed to provide additional margin and minimize the risk of spectral interference once routine imaging that requires leveraging the speed advantage of multi-frequency excitation commences.

Scanning along the fast-axis x is much faster than scanning along the slow-axis y , so AsyncOCE typically uses fewer samples along y to reduce acquisition time. As a result, amplitude modulation in the frequency domain manifests differently along each axis. For example, for typical AsyncOCE imaging parameters, such as $N_0 = 2048$ A-lines per B-scan, $N = 1920$ A-lines per B-scan after cropping, 2 kHz shear wave excitation, and 100 kHz A-line rate, modulation along the x -axis appears straightforward. As the shear wave frequency increases, the carrier frequency also increases linearly. The desired signal then shifts farther from $k_x = 0$.

The carrier frequency along the k_y axis also increases linearly. However, the time delay between y measurements is much longer, so the demodulation pixel aliases up to hundreds of times. With $M = 75$ sampling locations along y and $N_B = 2$ repeat B-scans per y_m , the demodulation pixel k_{y_0} can be solved using Eq. (4). This gives $k_{y_0} = 6114$. Because the data is discrete, this value aliases by modulus M to -6. This creates a seemingly random distribution of signals along the k_y axis, but it is fully deterministic. This aliasing is also advantageous because small changes in excitation frequency produce large shifts along k_y . This allows many more frequencies to be packed into k -space than might be expected.

2.4. Experimental setup and materials

To validate multi-frequency AsyncOCE, we constructed a bulk, homogeneous, aqueous, tissue-mimicking phantom composed of 2% (w/w) agarose and 0.5% (w/w) intralipid. We performed multi-frequency AsyncOCE measurements by simultaneously driving five shear wave frequencies (500, 800, 1200, 1845, and 2570 Hz). We also acquired conventional AsyncOCE data for validation by individually and sequentially driving each frequency.

To examine the impact of boundary conditions on multi-frequency shear wave measurements, we constructed a thin (1 mm), homogeneous, aqueous, tissue-mimicking phantom composed of 2% (w/w) agarose and 0.5% (w/w) intralipid. The same 5 frequencies (500, 800, 1200, 1845, and 2570 Hz) were used for shear wave excitation. The thin phantom was created from the same batch as the bulk phantom to guarantee the underlying material properties would be identical. During imaging, the thin layer rested on top of a petri dish with water, such that the upper boundary was in air while the lower boundary was in water, similar to the approach described in Pitre et al. [23]. Each sample was imaged three times and repositioned between acquisitions to a different region. We also imaged the homogeneous bulk sample with 12 frequencies (559, 744, 955, 1140, 1326, 1536, 1917, 2128, 2313, 2499, 2709, and 3090 Hz) to demonstrate limits on key imaging parameters, such as number of A-lines per B-scan.

A heterogeneous tissue-mimicking phantom was also constructed to demonstrate the ability of multi-frequency measurements to improve elastography image quality and contrast. The same 12 shear wave excitation frequencies were used. The sample background was composed of 2% (w/w) agarose and 0.5% (w/w) intralipid. After solidifying at room temperature, a rectangular region of the sample was removed from its center, into which a softer phantom mixture composed of 1% (w/w) agarose and 0.5% (w/w) intralipid was poured and also allowed to set at room temperature.

All excitation frequencies were carefully selected to avoid k -space interference while ensuring sufficient displacement between B-scans. Since the shear wave field is harmonic, measured displacement is zero if the B-scan rate is an exact multiple of the wave period. To avoid this, maximum displacement occurs when the B-scan rate equals a multiple plus one-half of the shear wave period. However, if all excitation frequencies are set to be exactly a multiple plus one-half of the shear wave period, their carrier frequencies along k_y all collapse to $k_{y0} = 0$. This wastes k -space and limits how many frequencies can be imaged at once. To avoid this, we first select frequencies that maximize displacement. We then slightly adjust alternating frequencies up or down to shift their carrier positions along k_y . As a result, multiple frequencies can be packed more densely in k -space without significantly reducing displacement amplitude. An example calculation is provided in the supplemental code with accompanying [Dataset 1](#) (Ref. [28]).

2.5. OCE system

The OCT data were acquired using a custom-built system, previously described by Cannon et al. [29]. In brief, the system used an optically k -clocked, phase-stable VCSEL wavelength-swept laser (Thorlabs, SL131090) with a nominal A-line rate of 100 kHz and a center wavelength of 1310 nm. Since precise knowledge of the A-line rate f_A is critical for AsyncOCE processing, it was measured directly. This was done by recording the auxiliary trigger output from the digitizer using a data acquisition board (National Instruments, PCI-GPIB N114) during background spectrum acquisition. The measured A-line rate was slightly different from nominal, at 100.084 kHz.

Since the A-line rate strongly affects shear wave demodulation, careful calibration was required. For an acquisition with a 2 kHz shear wave frequency and 1536 A-lines per B-scan across 150 B-scans, even a small error in f_A produces measurable phase drift. An error of 0.001 kHz (e.g., using 100.083 kHz instead of the measured value) leads to a cumulative phase offset of 0.29 radians over the volume, corresponding to 4.6% of a wave period. An error of 0.01 kHz (e.g., using 100.074 kHz) leads to a 2.89 radian offset, or 46% of a wave period. Using the nominal 100 kHz value results in a 24.3 radian offset over the full volume, equal to 3.87 wave periods. Such phase error would significantly distort shear wavenumber measurements across the entire field of view.

The collimated input beam had a diameter of 3.6 mm; combined with a 54 mm effective focal length objective scan lens (Thorlabs, LSM04), the system produced a resolution volume with an e^{-2} diameter of 25 μm after accounting for the confocal effect. Beam scanning was performed with galvanometers (Thorlabs, GVS002) over a 12 by 12 mm field of view, with alignment adjusted to minimize phase shifts during scanning. The interference fringes were detected with polarization-diverse receivers (Advanced Fiber Resources) and digitized using a 4GS/s digitizer with k -clocking (AlazarTech, ATS 9373). Shear waves were generated by a function generator driving a high-performance voltage amplifier (Micromechatronics Inc., MX200) connected to a piezoelectric actuator (Digi-Key 445-181631-ND). Shear waves were coupled into the sample via six 3D-printed prongs evenly spaced at a diameter of 20 mm [21].

During imaging, we acquired either $N_0 = 1024, 1536, 2048$, or 2560 A-lines per B-scan, including flyback and downtime. After cropping out the flyback region, $N = 896, 1408, 1920$, or 2432 A-lines per B-scan remained, respectively. Two B-scans per $N_y = 75$ out-of-plane y -locations were acquired, resulting in 150 total B-scans, for a total acquisition time of 1.54, 2.30, 3.07, and 3.84 seconds per multi-frequency volume, respectively. The lateral field of view remained 12 by 12 mm across all samples.

2.6. Signal processing implementation

See [27] for sample dataset and accompanying code. For s shear wave excitation frequencies, s parallel demodulation procedures were performed (see Fig. 3 for a graphic depiction). First,

AsyncOCE acquires the shear wave field at temporal intervals spanning many wave periods as the beam scans across the tissue, while MB-mode approaches repeatedly sample the same location without moving the imaging beam. As a result, phase differences are larger in AsyncOCE, and phase wrapping can occur. However, the OCT speckle pattern inherently provides a means of assessing spatial correspondence between measurements. In addition, established phase-unwrapping methods can be used to recover the underlying phase when phase wrapping occurs [30].

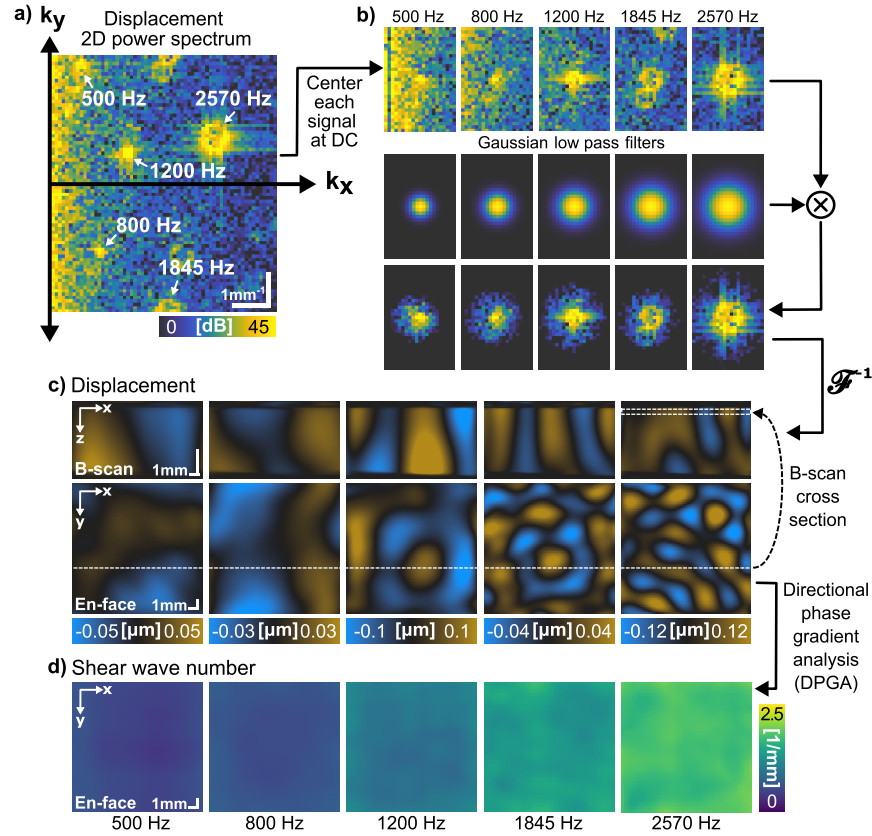


Fig. 3. (a) The displacement power spectrum along k_x and k_y is shown. Five distinct carrier frequencies are observed, alongside low frequency and harmonic noise. (b) After down shifting, each spectrum is low pass filtered with a 2D Gaussian filter. Filter widths were chosen to preserve all of the shear wave information from a given frequency while suppressing interference from other frequencies. (c) B-scan displacements may be observed by taking the real component of the displacement field. En-face displacement fields are shown after both in-plane and out-of-plane recovery are performed for each frequency. To restrict analysis to laterally propagating Rayleigh surface waves, only 200 μm from the surface was averaged along depth. (d) Finally, directional phase gradient analysis is performed to extract shear wave number from each displacement field.

Following phase unwrapping, the phase was converted to displacement using Doppler OCT [31]. Each spectrum was then shifted along both k_x and k_y so the carrier frequencies were centered on 0 Hz. They were shifted by applying a phase ramps along both x and y in the spatial domain. Each frequency required a unique shift depending on its carrier frequency.

Following Eq. (3), for $f_n = 500, 800, 1200, 1845$, and 2570 Hz, $f_A = 100.084$ kHz, and $N = 1920$ A-lines after cropping, the demodulation pixel values along k_x were 9.59, 15.35, 23.02,

35.39, and 49.30, respectively, which correspond to 115.0, 184.2, 276.2, 424.7, and 591.6 mm⁻¹. Following Eq. (4), with the same excitation frequencies, $N_B = 2$, $N_0 = 2048$ raw A-lines before cropping, and $M = 75$, the demodulation pixel values along k_y were 1534.7, 2455.6, 3683.3, 5663.1, and 7888.5 respectively. These signals aliased to pixel values along k_y of 34.7, -19.4, 8.3, -36.9, and 13.5 and correspond to 416.4, -232.8, 99.6, -442.8, and 162 mm⁻¹, respectively. The aliased pixel values were computed by $k_Y \bmod M$, where $k_Y \in [-M/2, M/2]$.

Upon applying both phase ramps, the 2D Fourier transform was computed so we could apply a low-pass filter to suppress the other frequencies and noise. In the case of $f_n = 500, 800, 1200, 1845$, and 2570 Hz and $R_x = R_y = 12$ mm, the standard deviation of the 2D Gaussian low-pass filters were 2, 2, 2.5, 3, and 3.5 pixels respectively. After applying the low-pass filter in the Fourier domain, the inverse Fourier transform was applied to recover the complex-valued shear wave field. As with conventional AsyncOCE, the real part of the resulting signal reveals the displacement field at one point in time. An additional phase shift may be applied uniformly to the entire field to allow for animation of the shear waves at any arbitrary point in time. Notably, the complex-valued, analytical displacement fields contain all of possible information in a harmonic shear wave field: acquiring more time points (via more B-scan repetitions) does not lead to more complete information about the harmonic shear wave field, except for a potential increase in signal to noise ratio. For this reason, we used 2 B-scan repetitions in all the experiments in this work, for which the time difference was chosen to maximize signal to noise ratio without requiring more repetitions. The analytical shear wave field can also be obtained from conventional MB-mode data by applying the Hilbert transform. Although, in principle, in MB-mode imaging only two A-lines per location are sufficient to recover the analytical shear wave field, the displacement magnitude during a time period of two consecutive A-lines is too small, making it necessary to acquire significantly more samples to increase signal to noise ratio [32].

The separated and recovered shear wave fields now had lateral dimensions of $[N, M] = [1792, 75]$. Because the x_n dimension is significantly oversampled after low-pass filtering, it was down-sampled to match the y -dimension and thus lower the computational requirements of subsequent processing steps. Restricting analysis to laterally-propagating Rayleigh surface waves allowed us to assume that shear waves near the surface may be averaged along depth, which we carried out for the top 50 pixels to collapse the $50 \times 75 \times 75$ volume into a 75×75 pixels 2D shear wave field. Then, we applied directional phase gradient analysis (DPGA) to extract a spatially-resolved shear wave number using non-overlapping window sizes of 2 by 2 pixels. The resulting image had dimensions of 37×37 pixels. From the shear wave number, we computed shear wave speed from

$$c_n = f_n / k_n, \quad (10)$$

where each frequency f_n had its own shear wave speed measurement.

We also implemented reverberant optical coherence elastography (RevOCE) for comparison in the heterogeneous sample. Although shear waves were recovered using AsyncOCE, the resulting displacement fields were analyzed using the RevOCE autocorrelation approach. Following Parker et al. [33], the 2D autocorrelation of the displacement field was computed in small windows along the x - and y -axes. Then, we took the normalized real part of the 2D autocorrelation, computed the azimuthal average, and fit the 1D result to the following sum of spherical Bessel functions

$$B_{S_z S_z}(\Delta \epsilon_{x,y}, \Delta t) = j_0(k \Delta \epsilon_{x,y}) - \frac{j_1(k \Delta \epsilon_{x,y})}{k \Delta \epsilon_{x,y}}, \quad (11)$$

where S_z is the displacement along z and $\Delta \epsilon$ represents the difference along a given dimension for which the autocorrelation is calculated. RevOCE was applied to the same down-sampled 75×75 pixel en face displacement field as DPGA. We used windows of 14×14 pixels to compute the autocorrelation. This was the smallest window size that yielded reasonably consistent results. We used the second zero crossing to fit the spherical Bessel functions. If the second zero crossing

was not within the maximum displacement difference $\Delta\epsilon_{x,y}$, we used the entire window to fit the spherical Bessel functions. Since the resulting shear wave maps from DPGA were 37×37 pixels, the resulting RevOCE map was up-sampled to match this size. We also applied a Gaussian smoothing filter with σ equal to 2 windows to the up-sampled RevOCE results.

2.7. Kelvin-Voigt fractional derivative (KVFD) model

Typically, in an isotropic and purely elastic material, wave speed is directly related to shear modulus G by $G = \rho c^2$ where ρ is material density and wave speed is constant across all frequencies. However, most biological tissues are not purely elastic, resulting in wave speed dispersion. The Kelvin-Voigt fractional derivative (KVFD) model is an extension of the Kelvin-Voigt model and has been shown to be a reasonable model of biological tissues with shear wave dispersion [34,35]. Since this model has already been described in detail, we summarize it here briefly. In the conventional Kelvin-Voigt model (KV), elasticity and viscosity are described using a spring and dashpot in parallel. The KVFD model modifies and extends the KV model by introducing a fractional-order element defined by a that can capture behaviors lying between ideal elasticity and ideal viscosity [see Fig. 6 (a)]. This fractional approach describes the gradual, power-law-type relaxation and creep typically observed in biological tissues [36].

First, the complex Young's modulus as a function of shear wave excitation frequency is

$$E^*(\omega) = E_s(\omega) + iE_l(\omega), \quad (12)$$

where $E_s(\omega)$ and $E_l(\omega)$ are the storage and loss moduli, respectively, and $\omega = 2\pi f$. Given the frequency-domain response of the KVFD model, and neglecting the elastic spring constant E_0 , the complex Young's modulus can be written as $E^*(\omega) = \eta(i\omega)^a$, which can be separated into its real and imaginary components:

$$E_s = \eta\omega^a \cos\left(\frac{a\pi}{2}\right), \quad (13)$$

$$E_l = \eta\omega^a \sin\left(\frac{a\pi}{2}\right). \quad (14)$$

Here, η is the fractional viscoelastic coefficient and a is the fractional-order parameter. When $a = 0$, the response is purely elastic, whereas when $a = 1$, the response is purely viscous. For a complex shear modulus $G^* = E^*/3$, the shear wave phase velocity is

$$c = \frac{|E^*|}{\sqrt{3\rho}} \sqrt{\frac{2}{|E^*| + E_s}}. \quad (15)$$

Substituting E_s and E_l into this expression allows η and a to be directly fit to the experimentally measured dispersion curves. Although the KVFD model has been extensively studied, boundary conditions are often simplified. This assumption is reasonable in bulk abdominal tissue, where boundary effects are small. However, in a thin plate, where the layer thickness is on the order of the shear wavelength, shear waves become guided and propagate as Lamb waves.

The classical Lamb wave formulation assumes a linearly elastic material. However, as discussed above, biological tissues are viscoelastic. To account for this, Urban et al. derived the dispersion relation for anti-symmetric Lamb waves in a Kelvin-Voigt material [37]. To ensure that the measured displacement was dominated by the anti-symmetric (A0) mode, displacement was averaged across the entire depth of the thin sample. For a homogeneous plate, the z -displacement of the symmetric (S0) mode is antisymmetric about the midplane, and opposing displacements cancel during depth averaging. Without depth averaging, contributions from the S0 mode may bias the measured shear wave speed toward artificially high values because the S0 mode

propagates faster than the Rayleigh wave speed. The dispersion relation is

$$4k_L^3\beta_l \cosh(k_L h) \sinh(\beta_l h) - (k_S^2 - 2k_L^2)^2 \sinh(k_L h) \cosh(\beta_l h) = k_S^4 \cosh(k_L h) \cosh(\beta_l h), \quad (16)$$

where k_L is the measured Lamb wavenumber, $\beta_l = \sqrt{k_L^2 - k_S^2}$, and $k_S = \omega\sqrt{\rho_m/G}$. For an isotropic, nearly incompressible material, the complex Young's and shear moduli are related by $E^*(\omega) = 3G^*(\omega)$. The objective is to estimate the complex shear modulus G^* , where

$$G^* = G_1 + i\omega G_2, \quad (17)$$

and G_1 and G_2 are the shear elasticity and viscosity, respectively. To relate this formulation to the Kelvin-Voigt model above for a bulk material, G_0 is neglected and the complex shear modulus G^* is expressed as

$$G^* = \frac{\eta}{3}(i\omega)^a. \quad (18)$$

2.8. Incoherent averaging of multi-frequency AsyncOCE shear wave number maps to improve image quality

Multi-frequency measurements are essential for characterizing viscoelastic tissues. However, most multi-frequency analyses report a single bulk value, implicitly assuming homogeneity. This assumption likely reflects the sensitivity of power-law fits to noise, which often requires averaging over large regions. In some applications, spatial resolution and contrast are more important than precise parameter estimates. For example, in tumor margin assessment, the heterogeneous nature of the disease makes high spatial resolution more critical than accurate global mechanical property values. Diffuse harmonic shear wave fields also contain low-displacement nulls, which produce regions of low signal. Combining data from multiple excitation frequencies can help mitigate these effects.

Several approaches for integrating multi-frequency measurements have been proposed in both magnetic resonance elastography and ultrasound elastography. However, most assume negligible viscoelastic dispersion and treat elasticity as constant across a measured frequency band. Since we expect non-negligible wave-speed dispersion, averaging measurements across frequencies would be inaccurate. Instead, we normalized and rescaled measurements across frequencies to account for such frequency-dependent shear wave speeds. Although this rescaling removes quantitative, viscoelastic information, yielding just a relative stiffness map ranging from 0 to 1, it enables more meaningful averaging across frequencies to preserve spatial contrast. Furthermore, we limited analysis to samples containing two distinct elastic regions, for which values are theoretically binary (0 or 1), to avoid complications associated with relative scaling.

To achieve consistent image scaling while mitigating the influence of outliers, we applied percentile-based normalization to each shear wave speed map. The lower and upper intensity bounds were defined by the 5th and 95th percentiles of shear wave speeds. Values below or above these thresholds were clipped, and the remaining intensities were linearly scaled to range from 0 to 1 prior to averaging. Averaging was also weighted by the mean displacement magnitude of each frequency across the entire field of view. For heterogeneous samples, percentile-based clipping and normalization to a [0, 1] scale allows spatial variations in shear wave speed to be visualized consistently across different frequencies. In contrast, for homogeneous samples, where substantial spatial variation is not expected, normalization relative to the mean or maximum shear wave speed may be more appropriate.

3. Results

3.1. Validation with sequential measurements

Displacement fields acquired from multi-frequency excitation (Fig. 4(a),b) are shown alongside sequentially measured displacement fields (Fig. 4(c)). Both acquisition protocols were performed

in the same tissue-mimicking phantom without repositioning the excitation device, ensuring that the wave fields should remain identical. Sequential measurements took 11.5 seconds across five frequencies, while the multi-frequency measurement took just 2.3 seconds. For the multi-frequency excitation dataset, subsequent analysis for shear wave extraction was also performed with direct demodulation (Fig. 4(a)) in comparison to split-axis demodulation (Fig. 4(b)), which was previously validated in Schmidt et al. [21].

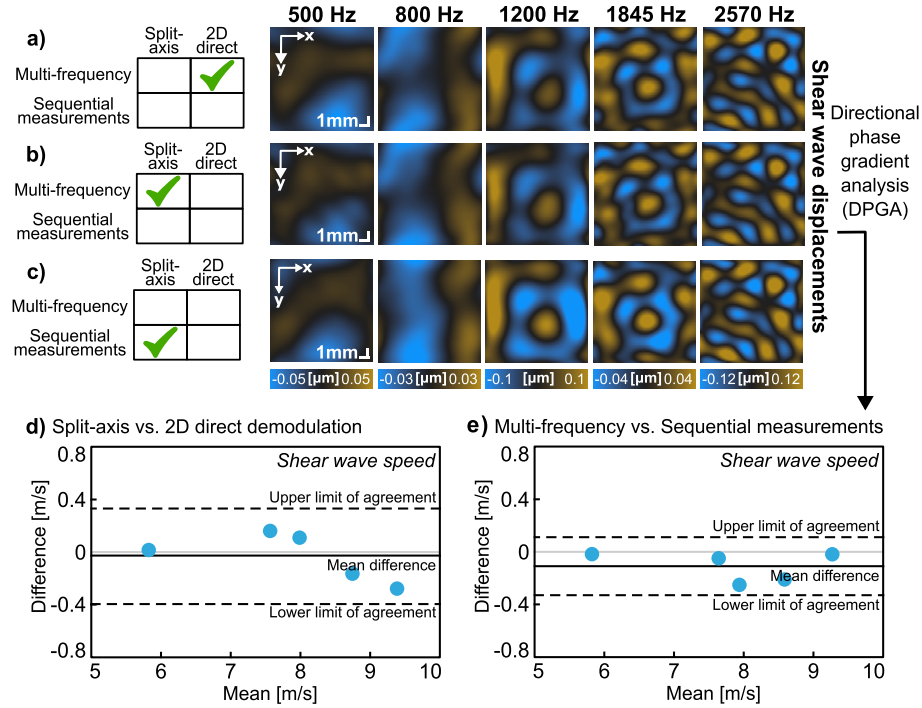


Fig. 4. (a) Displacement fields acquired via multi-frequency measurement with AsyncOCE and direct demodulation. (b) Displacement fields acquired via multi-frequency measurement with AsyncOCE and split-axis demodulation. (c) Displacement fields acquired sequentially via conventional measurement with AsyncOCE and split-axis demodulation. (d) Directional phase gradient analysis (DPGA) maps shear wave number k from the displacement fields. Bland-Altman analysis shows no statistically significant difference between direct and split-axis demodulation. The differences between the two techniques are plotted as a function of their mean, and the limits of agreement include zero difference, so they are not statistically distinguishable. (e) Similarly, there is no significant difference between measurements acquired simultaneously with multi-frequency direct demodulation versus sequentially measured with conventional AsyncOCE.

By inspection, all measurement and post-processing techniques recovered the same displacement fields. For a quantitative analysis, we applied directional phase gradient analysis (DPGA) to extract the shear wave number. We used Bland-Altman plots to compare the difference between the split-axis and direct demodulation as a function of their mean. We also compared multi-frequency with direct demodulation versus sequential measurements with split-axis demodulation, which was previously validated in Schmidt et al. [21]. In all cases, the shear wave speed did not show statistically significant differences. In both instances, the limits of agreement included zero difference. From this experiment, we conclude that multi-frequency AsyncOCE can reproduce the same shear wave fields as sequential measurements, provided that excitation frequencies and filter widths are chosen appropriately.

3.2. Predictable carrier frequencies and 2D direct demodulation conveniently enable multiplexing of many shear wave frequencies

We selected 12 optimized frequencies in the range [500, 3100] Hz for $N_0 = 1536$ A-lines per B-scan. Acquisition time was 2.3 seconds. After cropping out the galvanometer flyback region, $N = 1408$. These same 12 frequencies were subsequently applied to scans with $N_0 = 1024, 2048, 2560$ A-lines per B-scan, for which $N = 896, 1920, 2432$ A-lines respectively. Their acquisition times were 1.5, 3.1, and 3.8 seconds, respectively. The corresponding 2D power spectra are shown in Fig. 5, alongside the 1D, depth-dependent power spectra previously used for split-axis demodulation.

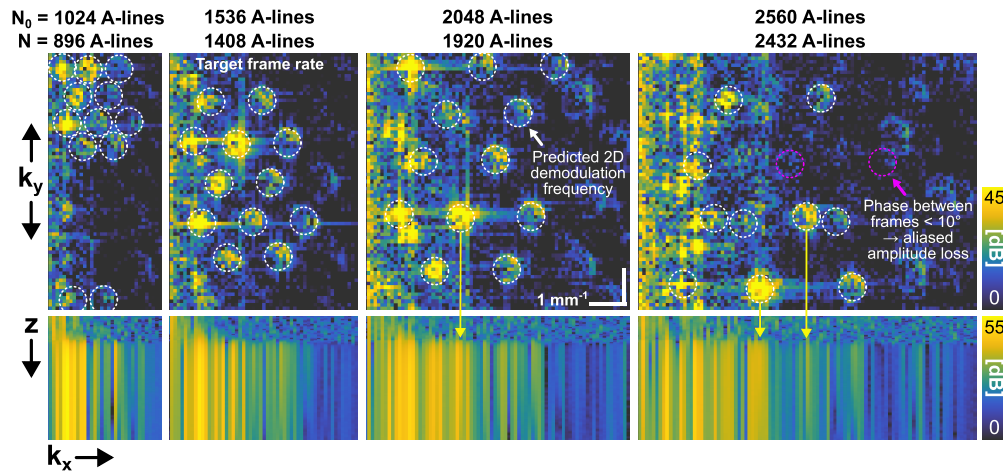


Fig. 5. 2D power spectra for 12 simultaneous excitation frequencies are shown in the top row, and the corresponding depth-resolved 1D power spectra along just the x -axis are shown in the bottom row. The previous AsyncOCE technique (split-axis demodulation) relied solely on the 1D spectrum, causing overlapping frequency components to appear inseparable. In contrast, the 2D spectrum enables straightforward isolation of individual excitation frequencies. The excitation frequencies were optimized for 1408 A-lines after cropping out the flyback region (second column). They were 559, 744, 955, 1140, 1326, 1536, 1917, 2128, 2313, 2499, 2709, and 3090 Hz. Signals corresponding to each frequency are highlighted in the white dotted-line circle. The white dotted-line circles were the theoretical *expected* locations for the carrier frequencies, and align nicely with the experimental results. At 2560 A-lines (2432 A-lines after cropping), two shear wave frequencies had a less than 10° phase difference between frames, resulting in aliased amplitude loss and low signal.

As the number of A-lines increases, the spread between signals also increases, as expected. The frequencies were chosen to maximize displacement measurements while maintaining sufficient separation from each other, analogous to preserving a near 180° shear wave phase difference between frames. Because the optimization was performed for 1536 A-lines, the signal is clearest for this case, and the shear wave signals are evenly spaced in the Fourier domain. For larger scans, such as 2560 A-lines, some frequencies produce shear wave phase differences below 10° , resulting in very small displacements and correspondingly low signal-to-noise ratios.

Importantly, the dotted circles indicate where we predicted the shear wave displacement would exist in k -space. We were able to accurately predict their carrier frequencies, even as imaging parameters such as the number of the A-lines per B-scan changed. We were also able to pack up to 12 frequencies all at once, potentially with room to increase to more frequencies. In the $N_0 = 2048$ case, there is at least one full ring diameter of frequency space between the edges of the bandpass filters at adjacent carrier frequencies. Therefore, we could have imaged a sample

with half the shear wave speed, which would increase the radius of each ring by a factor of two. If pathology is expected to be stiffer than healthy tissue, the resulting rings would be smaller, further reducing the risk of interference. Alternatively, this margin could be retained as a safety factor based on the expected range of shear wave speeds for a given anatomical target.

The 1D, depth-dependent power spectra previously used for split-axis demodulation also illustrate why interpreting multi-frequency results along a single axis is more challenging. This highlights the importance of direct demodulation as a tool for multi-frequency imaging.

3.3. Shear wave dispersion is sensitive to both viscosity and boundary conditions

To demonstrate the importance of multi-frequency measurements, data were also acquired in both bulk and thin tissue-mimicking phantoms. Shear wave excitation and imaging parameters were the same as in Section 3.1. It took 2.3 seconds to measure 5 shear wave excitation frequencies. Both phantoms were prepared from the same batch of agarose to minimize experimental variability, ensuring the material properties would remain as similar as possible. For the homogeneous thin sample, the displacement was averaged across the entire depth of the plate. The opposing S0-mode displacements cancel during depth averaging, while the A0-mode displacements are preserved. For the bulk phantom, the analysis focused on surface waves, which propagate laterally along the surface.

In the bulk sample, shown in blue in Fig. 6, shear wave speed increased with excitation frequency. Although bulk agarose phantoms are typically assumed to be purely elastic, the observed dispersion indicates the presence of viscous behavior. Measuring the shear wave speed at 1 kHz versus 3 kHz yields shear wave speeds that differ by a factor of about 1.5. Fitting the dispersion curve to the KVFD model yielded $\eta = 106.0$ Pa·s and $a = 0.68$, indicating that the bulk phantom exhibited intermediate behavior between purely viscous and purely elastic.

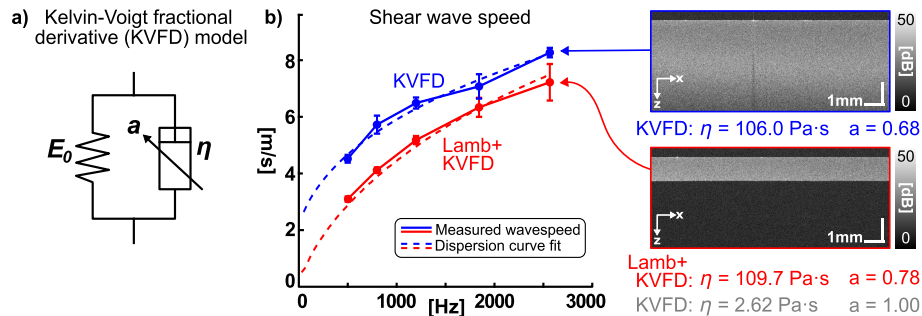


Fig. 6. (a) The Kelvin-Voigt fractional derivative (KVFD) model is an extension of the Kelvin-Voigt model where a spring (E_0) and dashpot (η) in parallel. The KVFD model, shown here, introduces a fractional-order element defined by a that can capture behaviors lying between ideal elasticity and ideal viscosity. (b) Shear wave speed dispersion curves for a thin-plate sample (red) and bulk sample (blue) are shown. Both phantoms were constructed from the same batch of agarose; however, boundary interactions in the thin plate strongly influence the measured dispersion curve. Dispersion curve fits are shown by dashed lines, while AsyncOCE multi-frequency measurements are shown by solid lines. KVFD fitting results are shown below the corresponding cross-sectional intensity images. For the thin plate, fits are shown using both the conventional bulk KVFD model (second row, gray) and a KVFD material model incorporating the Lamb wave solution for the A0 mode.

The thin phantom constructed from the same batch of agarose also displayed viscoelastic behavior; however its dispersion curve was different [see Fig. 6(b)]. Fitting it to bulk the KVFD model yielded $\eta = 2.62$ Pa·s and $a = 1.00$. These values are very different from the ones obtained from the bulk sample dispersion curve. Since the underlying material is identical, the observed

differences are due to unaccounted boundary conditions that create guided waves in the material, strongly influencing dispersion behavior [38]. When the Lamb wave model for a KVFD material was used to fit the experimental dispersion curve, $\eta = 109.7$ Pa·s and $a = 0.78$. These values are more consistent with the material properties measured in the bulk sample. The Lamb wave model is more complex, so the parameter estimates obtained from the bulk phantom using the KVFD model were used to initialize the optimization. Central to these results are the shear wave dispersion curve. Frequency-dependent information allows viscoelasticity and boundary conditions to be disentangled and provides a necessary framework for characterizing realistic biological tissues, such as in the cornea [23].

3.4. *Incoherently averaged and normalized multi-frequency measurements improve quality and contrast of qualitative elasticity imaging*

Figure 7 summarizes the stiffness reconstruction across excitation frequencies; results from DPGA and RevOCE are shown. The backscattered intensity image shows the softer, inner, rectangular region of interest, indicated by the dashed outline. However, from the structural image alone, we cannot distinguish which region is softer. The averaged stiffness map shows a clear contrast between the inner and surrounding regions, with higher values indicating greater stiffness.

Individual normalized shear wave speed maps are shown for all 12 excitation frequencies. While the overall spatial pattern is consistent across frequencies, higher-frequency maps exhibit increased noise. This is likely due to decreased shear wave displacement amplitudes at higher frequencies as a result of dispersion and attenuation, which reduce signal-to-noise ratio. By weighting the averaged stiffness map by the measured displacement magnitude, lower-confidence measurements, predominantly at higher frequencies, contribute less to the final estimate. This approach preserves spatial features that are consistently observed across frequencies while suppressing noise-dominated contributions, resulting in a more robust and interpretable stiffness map.

The improvement to image quality as a result of averaging across frequencies was quantified by the contrast to noise ratio (CNR)

$$C = \frac{|\mu_{\text{inc}} - \mu_{\text{bg}}|}{\sqrt{\sigma_{\text{inc}}^2 + \sigma_{\text{bg}}^2}}, \quad (19)$$

where μ_{inc} and μ_{bg} denote the mean signal intensity within the inclusion and background regions, respectively, while σ_{inc} and σ_{bg} denote the standard deviation of the inclusion and background signals, respectively. The frequency-averaged DPGA result exhibited a higher CNR (2.73) than RevOCE (1.85). This trend was also consistent across individual frequencies. Regardless of the shear wave process method, however, for both DPGA and RevOCE processing, the frequency-averaged results showed a clear improvement in CNR compared with any single-frequency measurement, demonstrating that combining elastography measurements across multiple frequencies enhances contrast.

4. Discussion

We introduced multi-frequency asynchronous optical coherence elastography, expanding the capabilities of shear wave elastography using conventional, clinically-available OCT systems. Building upon the strengths of AsyncOCE, which enables rapid acquisition of 3D shear wave fields in OCT with conventional A-line rates, multi-frequency AsyncOCE enables the determination of shear wave dispersion curves in the *same* acquisition time as a single frequency. This is a crucial capability because most clinical applications of OCE target tissues that are either viscoelastic or exhibit wave guiding due to boundary conditions, all of which are scenarios for

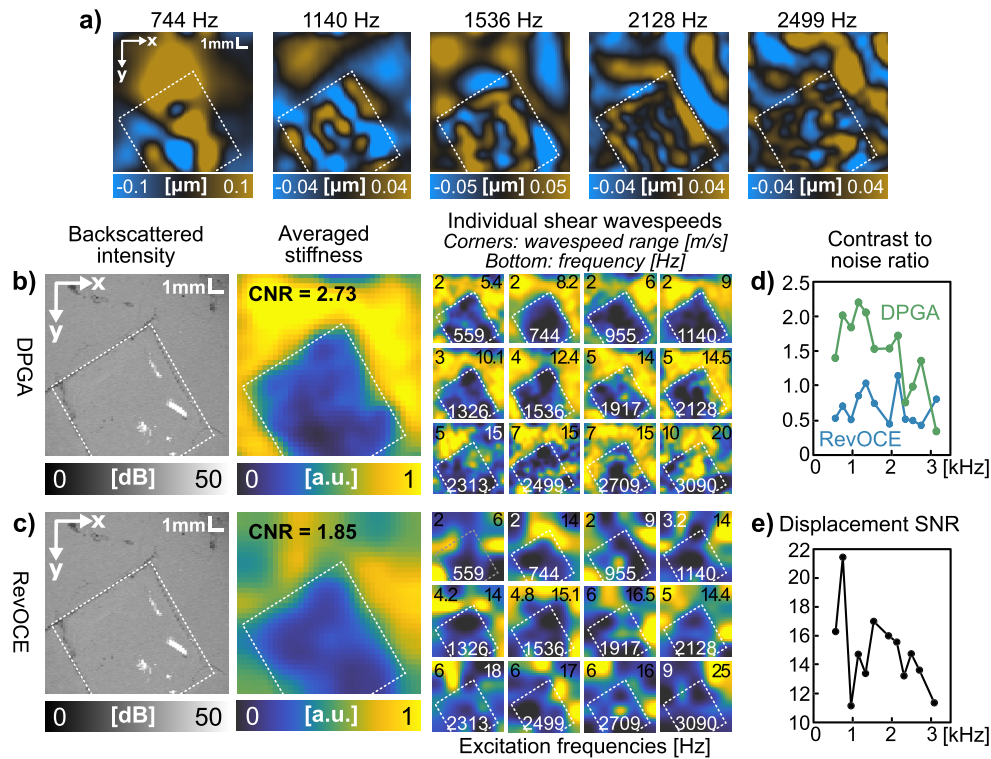


Fig. 7. (a) Select en-face shear wave fields from the surface of the 3D displacement volumes. (b) Elastography results from DPGA in heterogeneous sample. Backscattered OCT intensity image provides structural context, with the soft inclusion indicated by the dashed outline. Weighted, averaged stiffness map computed from all excitation frequencies shows clear mechanical contrast within the region of interest. Individual normalized shear wave speed maps at 12 excitation frequencies (559–3090 Hz) are also shown. While spatial patterns are largely consistent across frequencies, higher-frequency maps exhibit increased noise due to reduced shear wave displacement amplitudes. (c) The same results are shown for RevOCE processing. (d) CNR computed across individual frequencies, with comparison between DPGA and RevOCE. (e) Displacement SNR as a function of frequency.

which determination of shear wave velocity at a single frequency is insufficient for accurate model inversion to obtain elastic material properties. Although transient excitation OCE is capable of providing dispersion curves in a single excitation event, practical acquisition times due to MB-mode imaging have limited this modality to 2D shear wave imaging. This precludes its use in complex tissue where boundaries can create reflections with propagation directions that are not aligned with the imaging plane. Therefore, our development of multi-frequency AsyncOCE, which supports 3D shear wave imaging, further opens the door for quantifying the elasticity in tissue with complex boundary conditions, such as the cornea, or in tissues with heterogeneous composition, such as tumor margins or atherosclerotic plaques.

We demonstrated these concepts in phantoms with partially diffuse shear wave fields, in which we were able to determine their dispersion curves with a single, multi-frequency acquisition taking just 2.3 seconds. Meanwhile, heterogeneous samples also demonstrated improved contrast to noise ratio when multi-frequency fields were incoherently averaged. Similar to speckle reduction in OCT structural imaging, averaging mechanical information across frequencies enhances image quality in the presence of destructive interference or low displacement amplitudes. Importantly,

our multi-frequency framework enables this improvement without increasing the total acquisition time. This preserves the practicality of AsyncOCE in motion-sensitive environments *in vivo* such as when imaging through an OCT catheter. Physiological motion, including respiratory and cardiac motion, also occur at lower frequencies than the kilohertz-range shear wave excitations typically used in OCE. Furthermore, compared to other OCE methods, AsyncOCE modulates the shear wave displacement signal to an even higher carrier frequency through the raster scanning process, providing additional separation between the desired shear wave signal and motion artifacts. AsyncOCE scanning patterns are also less demanding than those typically used for angiography, requiring only two repeat B-scans rather than four. OCT angiography is already widely used clinically, so this similarity further supports the clinical applicability and simplicity of the AsyncOCE scanning approach.

There remains potential for further work. Our current approach for incoherent averaging of multi-frequency measurements treats each frequency independently and aggregates them through simple rescaling. This strategy improves contrast, but does not fully capitalize on the mechanical information encoded as a function of frequency. Future work is needed to develop principled approaches for combining frequency channels that adapts to local mechanical properties rather than applying uniform weighting, such as in Tzschätzsch et al. [24]. Furthermore, although this study focused on isotropic agar phantoms, biological tissues are often anisotropic and structurally heterogeneous. In such cases, the three-dimensional shear wave distribution recovered through direct demodulation may reveal directional dependence in stiffness or dissipation. Studies in clinically-relevant applications such as corneal biomechanics, tumor margin assessment, and cardiovascular tissue characterization are also promising.

5. Conclusion

In this work, we presented a new theoretical framework for multi-frequency AsyncOCE based on direct demodulation with the 2D Fourier transform. By leveraging the amplitude modulation induced by asynchronous scanning, we demonstrated that up to twelve shear wave frequencies may be simultaneously encoded within a single excitation event. We further evaluated the limits of multiplexing, identifying constraints on frequency density, interference, and acquisition parameters, and provided open-source code. Experimental validation confirmed that direct and split-axis demodulation yield equivalent wave number estimates and that multiplexed measurements maintain accuracy when compared to sequential acquisitions. We also demonstrated the importance of multi-frequency measurements by observing dispersion from both boundary reflections and viscoelasticity. We also showed that incoherent averaging across frequencies improves elastography image quality in heterogeneous media. Together, these results demonstrate multi-frequency AsyncOCE's potential as a scalable, high-throughput elastography modality capable of capturing complex mechanical behavior. We anticipate that this framework will substantially benefit studies *in vivo* where tissue heterogeneity, limited acquisition time, and complex wave fields demand richer spectral information.

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Data availability. Data may be obtained from the authors upon reasonable request. Sample processing scripts and test data are available at [Code 1](#) (Ref. [27]) and [Dataset 1](#) (Ref. [28]).

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